Catching Physical Hallucinations: A Causal Execution Layer for VLA and Autonomous Driving under Distribution Shift

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***Abstract*—Vision-language-action (VLA) policies are trained to produce actions, not to know when their actions stopped being true. Their one self-assessment is a scalar confidence, and a confidence is a function of the model while physical drift is a function of the world; the two can disagree without either being wrong. We describe a *causal execution layer* (CEL)—in one phrase, a *causal auditor* of the policy's own physical claims—that replaces the question "how sure am I?" with "did the world do what I said it would?". It never filters, clips or blocks an action: it audits the claim attached to one. Before each action chunk the layer registers the physical consequence its world model predicts; after the chunk it measures the consequence, quantises it in the plan's own frame, and offers it to a hash-linked gateway. A matching digest earns an *execution warranty*—evidence, bound to this attempt, that the commitment was redeemed. A refusal is written to the ledger and the plan is *re-derived* from the observed state rather than retried. On `libero_spatial` task 2 (200 episodes, two arms, identical policy) the layer raises success from 51.0% to 100.0%; under a 101–108 mm displacement of the target plate it is 100.0% against 0.0% ($p < 10^{-24}$); and the confidence baseline refuses 34 of 84 actions that would have succeeded. Total latency rises 1.7% while decision latency falls 19.3%, because the layer never pays for a blind retry.**

The layer is not specific to an arm. We move it to an autonomous-driving stack, where the drifted quantity is the friction of the road rather than the pose of an object and no camera observes it: on 100 episodes of a 280 m urban route, a confidence-gated planner ends 15 of 30 wet-road episodes in a pedestrian collision and the layer ends none (Fisher 5.8×10^{-6}); in a worn-steering channel where both arms reach the goal every time, 96% of the baseline's episodes arrive over the centre line against 0%. The same layer also recovers the physical parameters it was never given, $\hat{\mu} = 0.350$ and $\hat{\tau}_a = 0.400$ s, from its own saturated and stepped responses.

We also report a defect in our own first implementation: the gate was a set-membership test over all previously registered digests, so a scene that *regressed* to an earlier state was accepted as evidence that the action had worked. Three episodes passed every witness check with the manipulated object 245–269 mm from its target. A hash binds content; causation is a relation between a prediction and a particular attempt. Binding a monotonic ordinal into every registration closes the replay, and we argue this is a general requirement for any physical witness, not a bug fix.

***Index Terms*—vision-language-action models, runtime verification, physical grounding, causal interfaces, robot manipulation, distribution shift**

I. Introduction

The dominant interface between a language-level instruction and a torque-level action is now a single learned function: a vision-language-action model maps an image and a token sequence to a short action chunk [1], [2], [3]. Its evaluation, by construction, is nominal. LIBERO [4] and its harnesses measure success on the distribution the policy was trained on; when the distribution moves—a bumped table, a worn motor, a dimmed lamp—the policy does not notice, because nothing in it is a measurement of the world. It is a measurement of images.

The industry's answer is a confidence gate: act only if the policy reports a sufficient scalar, otherwise refuse or hand over [5], [6]. A more principled version of the same move is a safety filter or a control barrier function, which certifies the proposed input against a model of the dynamics [7], [8]; the newest VLA instantiation adds such a layer to a manipulation policy and reports large safety gains on a LIBERO variant [9]. These are real mechanisms, and they are the right baselines, but they share one assumption and therefore one blind spot. A confidence is $f(\text{model})$. Drift is $g(\text{world})$. When the world moves and the input distribution does not, f is entitled to be high and wrong. Conversely, when the input distribution moves but the physics has not—a dimmer light, a shadow—f is entitled to be low and right. The two failure modes are not opposite ends of one dial; they are different functions. A safety filter narrows the same gap without closing it: it is exact with respect to the model it is given, so a model that is wrong about the road is enforced faithfully.

This paper asks what changes if the gate is a *measurement of the consequence* instead of an opinion about the action. The layer is small. Before a chunk runs, its world model predicts the physical state at the end of that chunk and registers the prediction, canonicalised and hashed, with a hash-linked gateway. After the chunk, the *actual* physical state is read, canonicalised the same way, and offered as a witness. The gateway issues an execution warranty only for a digest that was registered before the request arrived. If it refuses, the ledger records the refusal and the layer re-plans from the

observation, resuming at the stage the world is actually in. The layer is therefore an auditor of physical truth and not a filter on action: it holds no safe set, states no constraint, and has no authority to clip a command—it decides only whether the claim that justified the command was true. The claim is narrow and falsifiable: *a witness that commits to a physical consequence distinguishes drift from uncertainty in a way a confidence cannot.*

a) Contributions:

1) A causal execution layer for chunked VLA rollouts, with the witness defined in the plan's own reference frame so that grid quantisation does not manufacture refusals at cell boundaries (Section III).
2) A controlled comparison on a real simulator: 200 episodes of `libero_spatial` task 2, two arms sharing one policy byte-for-byte, so the only difference is the gate. Success 51.0% vs 100.0%; under pose drift 0.0% vs 100.0%; 34 false vetoes against zero (Section V).
3) A latency decomposition that separates *decision* latency from *tamper-evidence* latency, and the negative result that our own chain verification is $O(\text{ledger})$: total latency is $+78$ ms ($+1.7\%$) while decision latency is -889 ms (-19.3%) (Section V-C).
4) A transfer of the same layer to autonomous driving, where the drifted quantity is a coefficient of friction instead of a pose and the outcome is a collision instead of a completed task: 15 collisions in 30 wet-road episodes against zero (Fisher 5.8×10^{-6}), and a worn-steering channel in which 96% of the baseline's successful episodes arrive over the centre line against none (Section VI).
5) A defect in the layer itself, reported rather than patched: the first version of the gate accepted a scene that had *regressed* to a state it had legitimately predicted earlier, because the gateway's registration set is not consumed. A hash binds content, not causation. The fix—a one-shot monotonic ordinal bound into every registration—is a general requirement for physical witnesses, and the reasoning is the strongest evidence we have for the paper's own thesis (Section VII).

The last item is deliberately in the abstract. A paper arguing that unbound witnesses are worthless should not be believed on the strength of its own witness without saying where that witness failed.

II. Related Work

a) VLA policies.: RT-2 [1], OpenVLA [2] and π_0 [3] established the image-to-action-chunk interface this paper takes as given. None of them carries a runtime check: the policy is a function, and safety is expected to come from the data or from a downstream filter. We take the interface as fixed and add the check.

b) Shielding, safety filters and control barrier functions.: Shielding [5] restricts an agent to a set of safe actions using a model of the environment; runtime verification [10], [11] checks a trace against a specification. The control-theoretic relatives are the predictive safety filter [7], which admits or minimally modifies a proposed input against a backup trajectory computed from a learned model, and the control barrier function [8], which enforces invariance of a safe set as a constraint on the input. These are the strongest mechanisms in the literature and the closest to ours in placement: they too sit between a policy and an actuator.

The difference is what they assume and what they measure. A CBF certifies safety *given* a safe set and a model of the dynamics; a predictive safety filter does the same given a model and its uncertainty bound. Stating such a specification at deployment is one of the concrete problems in safety identified a decade ago [12], and it is why both guarantees inherit the model the designer supplies. They are correct when the model is right, and the drift this paper is about is the case where it is not: a friction coefficient that changed, a steering rack that lags. The guarantee then degrades silently, because the quantity that moved is inside the model rather than in the observation. Our witness does not certify safety at all; it makes the model's error *observable* by comparing a registered prediction with a measured consequence, and then re-estimates from the consequence. CEL is not a shield, a safety layer or a safety filter: it has no safe set, certifies nothing, and never touches the action—it is a causal auditor of physical claims whose only outputs are an execution warranty and, when the claim is false, a refusal to issue one. A safety filter asks "is this action safe under my model?"; we ask "was my model true about this action?", which is answerable when the first is not.

c) Runtime guardrails for VLA.: The 2025 wave of VLA safety work is the nearest neighbour and deserves to be named. SafeVLA [6] folds safety into training as a constrained MDP; VLSA/AEGIS [9] adds a plug-and-play control-barrier layer that monitors a VLA's action in real time and intervenes only on predicted violations; RoboSafe [13] runs a guardrail model that emits executable safety predicates and re-plans when they are violated; AnySafe [14] synthesises latent safety filters inside a world model and calibrates the threshold conformally. All four are runtime interventions between the policy and the plant, and VLSA/AEGIS is evaluated on a LIBERO variant, which makes the comparison concrete rather than philosophical.

Two things separate this paper. First, each mechanism requires a constraint to be *stated*—an obstacle geometry, a set of unsafe predicates, a latent failure set—whereas CEL's specification is the policy's own next-state prediction, regenerated every chunk. Second, they modify or block the action to keep the system inside a set; we leave the action untouched, accept or refuse the *claim*, and re-derive the plan from the measured state. A guardrail that stops the robot when it is unsure is a veto; a witness that says "the bowl is on the table, so the transport stage is fiction" is a re-estimate, and Section V prices the difference in false vetoes. Nothing here diminishes the guardrail line: it is stronger than ours wherever a constraint can be written down, and weaker exactly where the world has moved and the constraint has not.

d) Uncertainty gating.: SafeVLA [6] aligns a VLA with safety constraints in training; the classic inference-time

version is a confidence threshold. We implement the inference-time version as arm A at its published threshold and price its refusals counterfactually, because a refusal rate alone cannot distinguish caution from blindness. The distribution-shift literature reaches the same conclusion from the other side: conformal guarantees computed on the training environment collapse when deployment differs, and the repairs either re-estimate the shift online [15] or model the uncertainty inside the planner. Both make a scalar carry information about the world; a witness replaces the scalar with a measurement.

e) Causal framing.: DoWhy [16] formalises the do-operator for observational data. Our witness is a do-test executed on a simulator: we do not infer $P(s' \mid \mathrm{do}(a))$, we measure it, once per chunk, and compare it with the model's prediction. The comparison is the causal claim. The closest work in this direction is CADET [17], which audits a deployed end-to-end driving planner for reliance on cues that cannot physically influence the decision, and repairs it by masking those queries at test time. CADET audits the *input* dependence of a plan with an offline physics-geometric prior; we audit the *outcome* of an executed plan against a prediction registered before it ran. An input audit cannot see a drift that leaves the input distribution intact—a wet road looks like a dry one—and an outcome audit can, which is the experiment of Section VI.

f) World models and physical hallucination.: Learned world models [18] are now the substrate for robot data generation, and their characteristic failure has a name: physical hallucination, in which generated rollouts interpenetrate objects or violate kinematics [19]. The remedy is verification against a simulator or a geometric prior [19], [20], which is available offline and not at execution time on a physical robot, where the only oracle left is the plant itself.

g) End-to-end driving.: UniAD [21] makes planning the objective of one network over a learned scene representation and VAD [22] vectorises the scene and samples trajectories from it; both are scored on completing a route without contact, on the distribution they were trained on. The output is a trajectory and a confidence, so the interface is unchanged—and the quantity that separates a safe plan from a fatal one, the friction of the surface, is not in the input. Section VI runs the comparison there.

III. The Layer

Fig. 1 fixes the vocabulary for the rest of the paper: a plan, a commitment registered before it is executed, a measurement after, and a gateway that either issues the warranty or refuses. Everything in this section is one of those four boxes.

A. Commitments, not opinions

A rollout is a sequence of chunks. At chunk k the policy observes s_k and emits an action chunk a_k (6 control steps, 0.3 s) together with the physical state it expects to see afterwards. The expectation is the model's, not ours; in our implementation it is a per-axis gain fitted once on nominal episodes (0.0115, 0.0132, 0.0129 m per unit command per step, residual 11.8 mm over 15 probes) and frozen before any drift is applied.

Definition 1 (Commitment)**.** For chunk k with observation s_k and action a_k, the commitment is the pair $C_k = \langle \hat{s}_{k+1},\, k \rangle$ where $\hat{s}_{k+1} = f(s_k, a_k)$ is the model's predicted physical state after a_k. The commitment is *registered* before a_k is executed.

The registration is what makes the check causal rather than post-hoc. A prediction written after the fact can always be fitted to what happened; a prediction written before, hashed, and appended to an append-only chain cannot.

The artifact a match returns is the *execution warranty*: a statement, verifiable later, that the commitment of one particular attempt was redeemed by the plant. It is a receipt for a claim, not a permission to act—the action in Arm B is the policy's, unmodified. This is the sense in which the layer audits rather than filters: an unmodified action with a checkable warranty is an audit trail, whereas a clipped action has lost the evidence of what the policy wanted.

B. The witness

The observed state is reduced to the part the plan can assert something about: the target object's pose and a three-valued classification of the manipulated object,

$$\text{bowl} \in \{\textsf{table},\ \textsf{gripped},\ \textsf{placed}\},$$

decided from geometry with a declared 20 mm lift threshold (a successful grasp lifts the bowl 85 mm; a missed one lifts nothing) and a declared 50 mm placement tolerance. The end effector is deliberately *absent* from the witness: the controller measures it every step, so including it would let the witness be right for the wrong reason.

a) Quantise the displacement, not the coordinate.: A digest over raw coordinates is unstable at cell boundaries: an object that happens to sit on a grid line flips its hash under sub-millimetre jitter, and the gate refuses for a reason that has nothing to do with the world. We therefore canonicalise in the plan's frame,

$$\kappa(s) = \left\lfloor \frac{s_{\text{plate}} - b_{\text{plate}}}{g} \right\rceil \parallel \text{claim}(s), \qquad g = 10\,\text{mm}, \tag{1}$$

where b is the plan's belief. Both sides quantise the same reference, so only the *displacement* is exposed to the grid. Ten millimetres is measured, not tuned: the plate and a resting bowl are predicted to sub-millimetre accuracy, but the plate itself settles by about 1 mm when a bowl is set down on it, and the smallest drift we inject is 80 mm. A 10 mm grid is five times the settling and an eighth of the signal.

C. Refusal is a re-estimate, not a retry

When the gateway refuses, the layer does not replay the chunk. It treats the pair $(\hat{s}_{k+1}, s_{k+1})$ as a measurement and asks a different question: *what stage is the world in?* The plan's grasp and placement targets are re-derived from the fresh observation, the per-axis gain is re-fitted from the prediction error, and the program resumes at the stage implied by the observed classification:

Observed	Resume at
gripped	wherever the plan was, if it was carrying
table	approach, i.e. grasp again
placed	retreat, i.e. let go and back off

This is the substantive difference from a confidence gate. A retry says "try the same thing again"; a re-derivation says "the bowl is on the table, so the transport stage is fiction, start from the pick". In our run the layer pays 94 re-derivations over 100 episodes, exactly one per refusal, and none of them is a repeat of a failed action.

D. The breaker

3 consecutive refusals, or 6 re-derivations in one episode, trip a circuit breaker and the episode is abandoned with a recorded reason. The breaker is not a safety mechanism; it is a *price* on the absence of progress. A plan that keeps paying forward passes for a world it cannot satisfy is the failure the layer exists to make visible.

IV. Experimental Protocol

a) Environment.: `libero_spatial` task 2—"pick up the black bowl from table center and place it on the plate"—through `OffScreenRenderEnv` on MuJoCo 3.2.3 [23] (pinned, not relaxed for this experiment) and robosuite 1.4.0 [24], rendering under EGL. Actions are the standard 7-DoF OSC delta pose at 20 Hz; a chunk is 6 steps, 0.3 s of committed motion. This is the interface every LIBERO policy emits, not a bespoke one.

b) One policy, two gates.: Both arms share a single chunked pick-and-place program: plan targets once from the first observation, commit to six-step chunks, servo within a chunk from proprioception. The policy is open-loop *within* a chunk on purpose—that is what a VLA is—so the comparison cannot be explained by feedback control that re-reads the world at every step. The arms differ only in what sits between the policy and the actuator.

c) Arm A (confidence-gated baseline).: Veto any chunk whose confidence is below 0.80, retry up to three times, then execute anyway: a VLA cannot skip its own action chunk, so a gate without an abandon path converts uncertainty into latency rather than into safety. The confidence signal is the model of a policy head described in Section VIII; its input is the mean brightness of the delivered frame, because that is the quantity a confidence head actually degrades with.

d) Arm B (CEL).: As in Section III, with a `MembraneGateway` issuing execution warranties against pre-registered digests and writing every grant and denial to a hash-linked ledger.

e) False vetoes are priced, not assumed.: A veto is counted as false only if replaying the vetoed chunk on a snapshot of the simulator advances the task or brings the bowl measurably closer to the plate. The replay is bookkeeping: it restores the snapshot and the episode's step, contact and drop counters, so the counterfactual never contaminates the run it scores.

f) Drift.: Five channels, all applied to the MuJoCo model and never to the observation: *pose drift* (the plate is displaced at the transport boundary), *surface slip* (all sliding frictions to 0.22), *payload* (bowl mass $\times 12$), *light* (headlight to 0.30), and *actuator* (`actuator_gainprm` to 0.60). The schedule is an explicit list, identical for both arms: 30 nominal episodes, 45 under pose drift in two blocks, then the four physical channels. Table II splits the results the same way.

g) The drift displacements are measured, not chosen.: This matters more than it looks. Two constraints pin the three pose-drift vectors. First, the arm's steady-state error grows to 10 mm to 100 mm beyond $x \approx 0.03\,\mathrm{m}$, $y \approx 0.30\,\mathrm{m}$, so a plate bumped there is outside the precise workspace and *no* planner can place on it; an experiment that bumps it there measures the arm's reach, not the layer. Second, the plate must stay clear of the neighbouring bowl at $(0.009, 0.306)$ m: a displacement that parks the plate inside that footprint leaves a gap too narrow for a 94 mm bowl, and LIBERO's `on` predicate—which requires actual contact—becomes unsatisfiable for any policy. All 3 vectors move the plate 101–108 mm through the region the arm tracks to sub-2 mm. We found both failure modes by running them; they are recorded because a benchmark that silently contains unsatisfiable episodes is not measuring the interface.

h) The latency ledger is arithmetic, and can be re-priced.: Two costs are declared rather than measured—one forward pass $c_f = 200$ ms, one confidence gate $c_g = 40$ ms—and everything else is measured. With N chunks, R retries, P re-derivations, W witness time and E episodes,

$$\begin{aligned} L_A &= \frac{N_A(c_f + c_g) + R_A c_f}{E}, \\ L_B &= \frac{(N_B + P_B)\, c_f}{E} + W_B. \end{aligned} \tag{2}$$

Substituting $N_A = 1827$, $R_A = 112$, $N_B = 1766$, $P_B = 94$, $W_B = 967$ ms/episode reproduces the reported means to the millisecond, which means a reader who disagrees with c_f can re-price both arms without re-running anything.

i) Statistics.: Proportions carry 95 % Wilson intervals, which unlike the normal approximation stay inside $[0, 1]$ at the four cells that are 0 or n. The two arm-level comparisons are Fisher exact tests, for the same reason.

V. Results

A. Success

Table I is the headline, and Fig. 2(a) is the same numbers as bars so that the shape of the result is visible before the intervals are read. At nominal, both arms are perfect: 100.0% and 100.0%, over 30 episodes each. That equality is the control—if the layer cost accuracy on the easy world, nothing below would be interpretable.

Under pose drift the arms separate completely. Arm A scores 0.0% [0.0, 7.9] and arm B 100.0% [92.1, 100.0] over 45 episodes, a difference of 100.0 points (Fisher exact $p < 10^{-24}$).

The mechanism is banal once stated. The plan froze a placement target when it first looked at the scene; the plate then moved 101–108 mm. Arm A executes its stale target faithfully

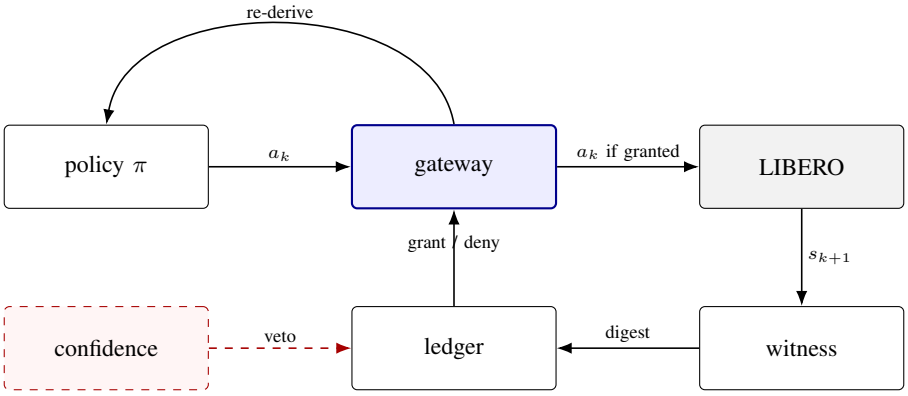


Fig. 1. The layer. Arm B routes every action chunk a_k through a gateway that issues an execution warranty only for a physical consequence registered *before* the chunk ran; the state s_{k+1} is read after the chunk, canonicalised by the witness, and offered to the ledger. A refusal re-derives the plan from s_{k+1}. Arm A (dashed) replaces the witness with a threshold on the policy's own confidence, so its only input is the model. Both arms drive the same simulator with the same action interface; the gate is the entire difference.

TABLE I

THE TWO ARMS ON `LIBERO_SPATIAL` TASK 2, 100 EPISODES EACH. BRACKETS ARE 95 % WILSON INTERVALS (NOT REPORTED BY THE HARNESS; COMPUTED BY `MAKE_NUMBERS.PY`). LATENCY IS PER COMPLETED EPISODE AND INCLUDES THE ARM'S OWN RETRIES; THE DECOMPOSITION IN TABLE III IS THE ONE TO READ.

	Conf.-gated	**CEL**	**Diff. (pts)**
Success			
Success, all episodes (n=100)	51.0 [41.3–60.6]	100.0 [96.3–100.0]	+49.0
Success, nominal (n=30)	100.0 [88.6–100.0]	100.0 [88.6–100.0]	0
Success, pose drift (n=45)	0.0 [0.0–7.9]	100.0 [92.1–100.0]	+100.0
Success, physical channels (n=25)	84.0 [65.3–93.6]	100.0 [86.7–100.0]	+16.0
What the gate did			
Vetoes / retries	84 / 112	0 / 0	
Re-derivations from a witness	0	94	
False vetoes / refusals	34/84 = 40.5%	0	
Episodes to recover after onset	27, 22	0, 0	
Physical cost			
Contacts (total / per episode)	68 / 0.68	9 / 0.09	
Episodes with damage	8 / 100	7 / 100	
Worst final bowl–plate gap	192 mm	29 mm	
Latency per episode			
Total policy latency	4609 ms	4687 ms	
of which tamper-evidence	0 ms	967 ms	
of which decision	4609 ms	3720 ms	

TABLE II

THE SAME 100 EPISODES, SPLIT BY THE WORLD THEY RAN IN. *False* COUNTS VETOES THAT THE COUNTERFACTUAL REPLAY SHOWS WOULD HAVE ADVANCED THE TASK; *ref.* COUNTS CEL'S WITNESS REFUSALS, EVERY ONE OF WHICH THE LEDGER RECORDS AS A MEASURED MISMATCH.

World	n	Success		Gate	
		Safe	CEL	veto (false)	refusals
nominal	30	100.0	100.0	10 (5)	11
pose drift	45	0.0	100.0	19 (5)	57
surface slip ($\mu = 0.22$)	7	100.0	100.0	2 (1)	6
payload ($\times 12$)	6	100.0	100.0	3 (0)	9
light ($\times 0.30$)	6	83.3	100.0	38 (18)	4
actuator ($\times 0.60$)	6	50.0	100.0	12 (5)	7
All	100	51.0	100.0	84 (34)	94

TABLE III

THE WITNESS'S OWN PRICE, MEASURED ON A SYNTHETIC LEDGER OF THE SAME SHAPE. EVERY GRANT RE-VERIFIES THE WHOLE HASH CHAIN BEFORE IT IS ISSUED, SO THIS COST IS LINEAR IN THE LEDGER AND THE RUN PAYS MORE AT THE END THAN AT THE START. THE SCENE CANONICALISATION AND HASH – THE ACTUAL WITNESS – IS 0.77 µs.

Ledger entries	Chain verify	Grant + redeem
202	2.2 ms	6.5 ms
1002	10.1 ms	18.5 ms
3002	31.0 ms	47.0 ms

and sets the bowl down on empty table, finishing 113 mm from the plate on average. Its confidence is not low: by the time the placement chunks run, the scene looks normal, the policy's input is unremarkable, and there is nothing for a head trained

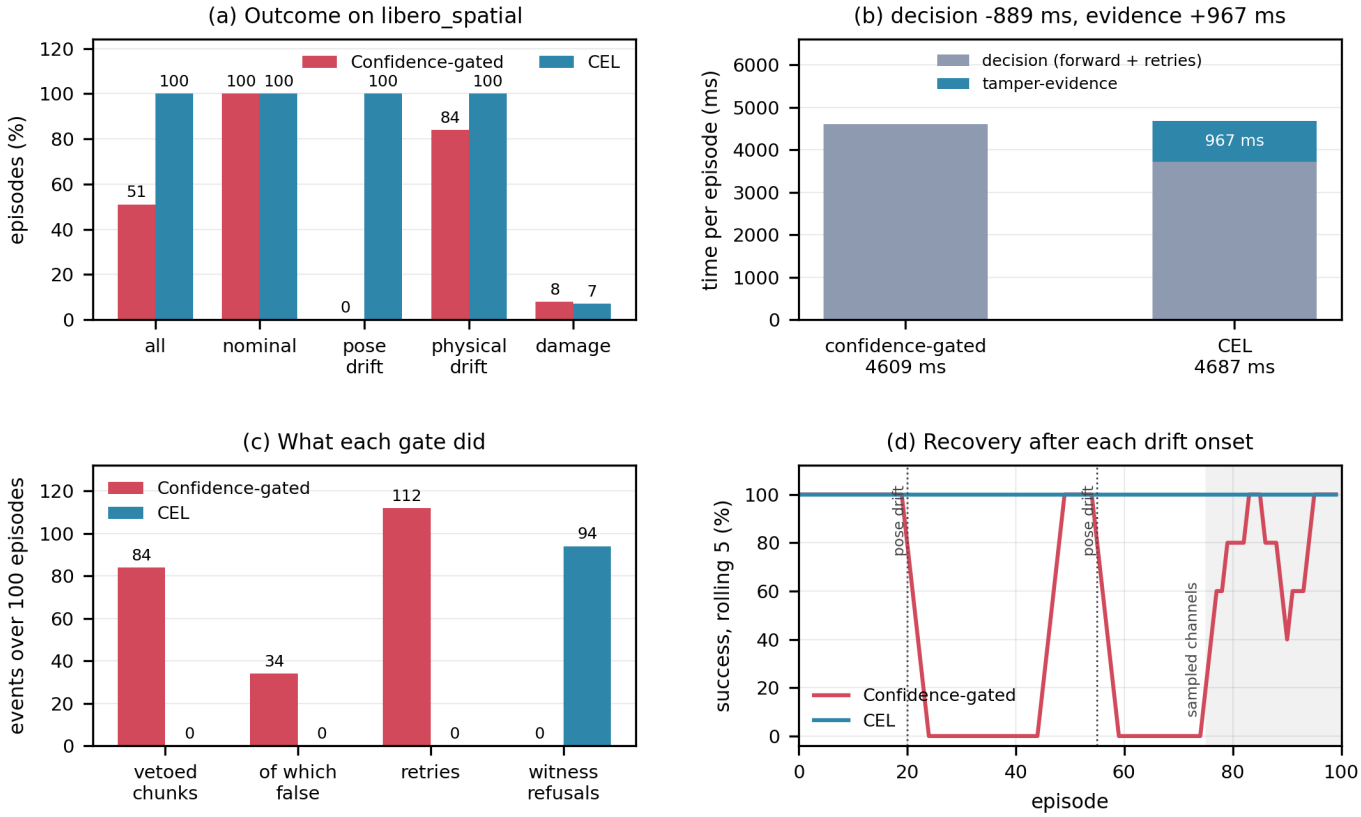


Fig. 2. The manipulation audit, all four panels drawn from the run's own episode records. (a) Outcome rates on `libero_spatial`: overall success, success in the nominal and pose-drift regimes, success over the drift channels, and the damage rate. (b) Time per episode, split into the decision (forward passes and retries) and the tamper-evidence; the split is the point, not the total, and the axis carries both parts so the total can be read off. (c) What each gate did over the hundred episodes—the baseline's vetoes, how many of them were false, and the retries they cost, against the layer's refusals. The baseline cannot refuse at all, which is why its last bar is empty rather than small. (d) Success over a rolling window of five episodes, against the two pose-drift onsets: the layer recovers inside the onset episode, the baseline never does.

on nominal data to notice. Arm B sees the plate's displacement at the first transport chunk, refuses, re-derives targets from the new pose, and finishes 6 mm from the plate.

Recovery tells the same story per onset, and Fig. 2(d) shows it per episode rather than as two numbers. Measuring episodes until the rolling three-episode success rate returns to 0.8: arm A recovers after 27 and 22 episodes; arm B after 0 and 0. A baseline that never notices the world moved cannot recover from it; the two numbers are not a difference of degree.

On the four physical channels arm A scores 84.0% and arm B 100.0%. The honest reading is that those channels are *easier* than a 100 mm plate displacement—Table II shows why: a 12-fold payload and a 0.22 friction coefficient mostly survive a nominal grasp (measured lift 112 mm and 112 mm against 115 mm at nominal), and *light* is inert for a policy that reads privileged state. Only the weakened actuator changes the arm as such (60 mm, 53% of nominal). We report them as channels, not as headline, and Table II reports each separately rather than pooling five phenomena into one average.

B. *The baseline refuses the wrong things*

Arm A vetoed 84 chunks over 100 episodes, and 34 of those refusals—40.5% [30.6, 51.2]—would have advanced the task. That is the metric a confidence gate is never scored on, and it is the one that decides whether the mechanism is usable.

The distribution in Fig. 2(c) is what makes the case that the threshold is not tunable. 38 of the 84 vetoes occur in the six *light* episodes, 18 of them false: the frame got dark, so the head got unsure, so the arm stopped doing something that would have worked. A further 5 false vetoes occur at *nominal*—where by construction nothing has changed. Lower the threshold and the light episodes stop being vetoed, but so do the drift episodes, because the signal is one scalar driven by the frame; there is no value of it that separates "the world moved" from "the lamp dimmed".

The layer has the opposite profile. Its 94 refusals are all measured mismatches between a registered prediction and an observed state, and 57 of them occur under pose drift. Its false-refusal count is zero by construction, not by luck: a refusal means the digest the model committed to was not the digest the world produced. When it is wrong about the world, it is

wrong in the direction the ledger records.

C. Latency: the layer is cheaper where it decides

Total latency per episode is 4609 ms for arm A and 4687 ms for arm B (Fig. 2(b), which also shows how the total divides): the layer costs 78 ms, 1.7% more. Taken alone, that number would be a fair reason to reject the layer, and we do not hide it. But the total is the sum of two things that are not alike.

Decision latency—forward passes, gate reads, retries, re-derivations, the work the policy does to choose an action—is 4609 ms for arm A and 3720 ms for arm B. The layer is 889 ms cheaper, 19.3% less, because it never pays for a blind retry: arm A spends 112 forward passes on retries that end by executing anyway, while arm B's 94 re-derivations are the whole of its extra work and every one of them is a response to a measured failure. *Tamper-evidence* latency—the cost of proving the ledger was not rewritten—is 967 ms per episode for arm B and zero for arm A by construction. Equation (2) shows the split exactly.

a) The tamper-evidence term is a defect of ours, and it is super-linear in the useful sense.: Every grant re-verifies the entire hash chain before it is issued, so the price of a witness grows with the length of the ledger that witnesses it. Table III: 2.2 ms at 202 entries, 10.1 ms at 1002, 31.0 ms at 3002—a factor of 14. The witness itself, canonicalisation plus SHA-256, is 0.77 µs. Over the run the per-chunk cost is 54.8 ms on average, 9.9 ms over the first ten episodes and 99.8 ms over the last ten; the growth is entirely the chain check.

This is an implementation defect with a known shape, not a property of the interface: an incremental accumulator or a Merkle root anchored periodically makes verification $O(\log L)$ or $O(1)$ per grant, at the cost of a weaker guarantee between anchors. We state the price curve rather than the fix because the curve is what a deployment has to budget, and because a paper that reports only the flattering decomposition would be doing what arm A does.

D. Damage, and what the layer does not prevent

Contacts fall from 68 (arm A) to 9 (arm B): 0.68 against 0.09 per episode. Damage episodes are 8 versus 7—nearly equal, and the reason is reported rather than smoothed: both arms drop the bowl four times (4 and 4), because a slip that happens during a committed chunk is a slip that has already happened when the witness sees it. A post-hoc witness cannot un-drop an object. What it can do is refuse to continue placing a bowl it is no longer holding, and that is the difference visible in the worst case: the largest final bowl–plate gap is 192 mm for arm A and 29 mm for arm B. The layer bounds the *consequence* of a failure it does not prevent.

VI. Generalization to Autonomous Driving

A. A trajectory is the same commitment

An end-to-end driving planner does not emit a delta pose; it emits a trajectory and a confidence. UniAD [21] plans through a learned scene representation and VAD [22] vectorises the scene and samples plans; both are scored on whether the ego vehicle completes a route without contact, on the distribution they were trained on. The interface question is the one Section III asks, and in driving the blind spot is sharper than in manipulation. Whether a pedestrian 30 m ahead can be avoided is decided by the coefficient of friction of the road, and no camera observes it: a wet surface differs from a dry one in reflectance, not in geometry, and the encoder has seen both. The literature has attacked this from the planning side, with conformal guarantees re-estimated online when the environment shifts [15], and from the audit side, by measuring which scene elements a deployed planner's decision actually depends on [17]. Both are statements about the plan and its inputs. Neither measures what the vehicle did. A confidence head is therefore entitled to be confident on a road it cannot stop on. The failure is not that the model is wrong about the picture; it is that the picture does not contain the quantity the plan depends on.

B. Plant, route, and the two arms

The simulator is replaced by the smallest model that contains the physics the gate is blind to. The ego vehicle is a kinematic bicycle with a friction circle: braking is capped at μg and curvature at $\mu g/v^2$, so a wet road costs stopping distance quadratically and cornering speed only linearly. Actuators are first-order lags, $\dot{a} = (a_{\mathrm{cmd}} - a)/\tau_a$ and $\dot{\kappa} = (\kappa_{\mathrm{cmd}} - \kappa)/\tau_s$, which is what a worn rack is. The route is 280 m of urban road with a queue at a red light, a curve, a lead vehicle that either brakes at 8.0 m/s^2 or turns off, and a pedestrian who steps out from behind parked vans when the ego is 30–36 m away. The planner rolls out the *nominal* plant 2.5 s ahead at 2 Hz and reports a confidence. Drift is applied to the plant and never to that belief, so the rollout becomes wrong while the policy stays the one from the control condition.

Arm A is the confidence gate: below 0.80 it commands a 6.0 m/s^2 stop, otherwise it executes its trajectory blind. Arm B is the layer of Section III, transposed. At 10 Hz it registers the plan's next waypoint as a tuple (x, y, v) on the plan's grid, quantises the observed tuple in the same frame, and grants execution only while the discrepancy stays inside 1.0 m longitudinally, 0.5 m laterally and 0.5 m/s in speed; before each re-plan it also demands that the road remaining is at least the stopping distance implied by the grip it has *measured*. A refusal re-plans from the observed state with a jerk-limited take-over (6 m/s^3) written *inside* the new plan. That last clause is the lesson of Section VII one layer down: a take-over blended outside the plan contradicts the waypoints the plan has just registered, refuses itself, and cascades into a refusal loop.

Four worlds on one declared schedule, 100 episodes per arm, identical for both arms: nominal; wet ($\mu = 0.35$ against a believed 0.90); worn actuation ($\tau_s = 0.75$ s and $\tau_a = 0.40$ s against 0.08 and 0.10); and both together. The hazard is either the lead vehicle or the pedestrian, alternating within every channel, so a channel is not also a hazard.

TABLE IV

THE DRIVING AUDIT: 100 EPISODES PER ARM, ONE DECLARED SCHEDULE, IDENTICAL PLANNER AND ROUTE. *Coll.* COUNTS EPISODES ENDING IN CONTACT WITH THE LEAD VEHICLE OR THE PEDESTRIAN; *cross-track p90* IS THE 90TH PERCENTILE OF AN EPISODE'S LARGEST LATERAL DEVIATION FROM THE LANE CENTRE; *centre line* COUNTS EPISODES THAT PUT A WHEEL OVER IT (HALF-WIDTH 1.75 M). THE LAST TWO COLUMNS EXIST BECAUSE THE SUCCESS COLUMN CANNOT SEE THE FAILURE THEY COUNT: IN THE WORN-STEERING CHANNEL EVERY BASELINE EPISODE *arrives*, AND 96% OF THEM PUT A WHEEL OVER THE CENTRE LINE ON THE WAY. *Conf.* IS THE CONFIDENCE-GATED BASELINE OF SECTION IV.

		Collisions		**Success (%)**		**Cross-track p90 (m)**		**Centre line (%)**	
World	n	**Conf.**	**CEL**	**Conf.**	**CEL**	**Conf.**	**CEL**	**Conf.**	**CEL**
nominal	34	0	0	100	100	0.17	0.17	0	0
wet road ($\mu = 0.35$)	30	15	0	50	100	0.14	0.12	0	0
worn steering ($\tau_s = 0.75\,\mathrm{s}$)	26	0	0	100	100	2.75	0.67	96	0
wet + worn	10	5	0	50	100	1.48	0.41	10	0
all	100	20	0	80	100	2.43	0.66	26	0

C. Where the two are indistinguishable, and where they are not

In the nominal world the two arms are the same instrument (Fig. 3(a), first pair of bars): zero collisions, 100.0% and 100.0% arrivals, and 0.17 m of cross-track error for both. That equality is the control; if the layer cost accuracy on the easy world, nothing below would be interpretable.

On the wet road the gate loses half its episodes (Fig. 3(a)): success 50.0% against 100.0%, i.e. 15 collisions in 30 against 0 for the layer (Fisher 5.8×10^{-6}), with every one of those crashes a pedestrian at 7.2 m/s mean impact speed and the mean gap to the lead falling from 11.1 m to 5.7 m. Panel (b) is the same count split by what was hit. The mechanism is arithmetic, and the audit computes it rather than asserting it. At the instant the pedestrian becomes visible, a friction-limited stop fits inside the road available with 0.5 m to spare on average, and the per-episode deficit ranges from -2.2 (room to stop) to $+2.7$ m (already short). That average margin is smaller than the distance the car covers while its planner reacts: the tracker's own 0.20 s time constant is worth 2.8 m at the 13.8 m/s it was doing, and the largest margin anywhere in the channel is 2.2 m, so the latency alone eats it in every one of the 15 episodes even where a friction-limited stop is calculated to fit. On dry tarmac the same measurement leaves 15.7 m, several times the distance the latency costs. Wet tyres and worn actuators put the instant itself at $+5.0\,\mathrm{m}$: short before anything has moved. Panel (c) is that arithmetic in a single episode, the same seed for both arms: the gate is at 13.8 m/s when the pedestrian appears and is still at 7.3 m/s where she is.

The gate is not simply asleep. In 12 of the 15 wet pedestrian episodes it never fires, because nothing in the frame is wrong; in the remaining 3 it fires and the car collides anyway, because a 6.0 m/s^2 command on a 0.35 road saturates at $\mu g = 3.4\,\mathrm{m/s^2}$. The hardest stop the gate can ask for is a policy constant, while the deceleration actually required is a property of the surface, and the gate owns no measurement of the surface to compare its constant against. This is the structural point in its purest form: the deficit is not a threshold that was set too high, it is a quantity that is not in the input.

D. The failure a success rate cannot see

The worn-steering channel is the sharper result because every episode succeeds. Both arms arrive in all 26 episodes. The baseline's cross-track error at the 90th percentile is 2.75 m, more than the 1.75 m half-width of the lane, and 96% of its episodes put a wheel over the centre line, against 0% for the layer (Fisher 1.1×10^{-13}). A plan tracked through a lagging rack oscillates across the lane and still reaches the goal, so a benchmark that scores arrival rewards it.

The layer's 0.67 m comes from a predictor rather than a stiffer gain. Having measured τ_s, it aims the pursuit from the state the car will be in $\hat{\tau}_s$ seconds from now, which cancels a pure delay that no feedback gain can cancel without buying the oscillation it is trying to remove.

E. Identification without privileged state

Nothing in arm B reads μ, τ_a or τ_s; all three are recovered from the vehicle's own response. Grip comes from saturation: when the commanded deceleration has been flat for twelve control steps and the realised value is both large and below the command, the realised value *is* the friction limit, so $\hat{\mu} = |a|/g$. Lag comes from the step response, by inverting the first-order solution over the same window. Medians over the run, and Fig. 3(d) puts them beside the injected values: $\hat{\mu} = 0.350$ against 0.35 true, $\hat{\tau}_a = 0.400$ against 0.40, $\hat{\tau}_s = 0.742$ against 0.75 (range 0.722–0.843), and $\hat{\mu} = 0.900$ on the dry episodes. The layer then re-plans with the identified parameters, which is why its stopping-distance witness is a statement about this road rather than about a nominal one.

The estimator has a failure mode we hit and fixed. A first-order lag also makes $|a| < |a_{\mathrm{cmd}}|$ for a few steps, so a braking transient looks exactly like saturation; the first version therefore reported a wet road on dry tarmac whenever it braked hard. Requiring a steady-state window before a deficit counts as saturation removed it, and `tests/test_ad_cel_audit.py` pins the property on six nominal episodes.

F. What the layer costs

A confidence gate spends comfort to buy time; the layer spends time to buy comfort. Over 100 episodes per arm the baseline spends 256 hard-braked events against 177 for the

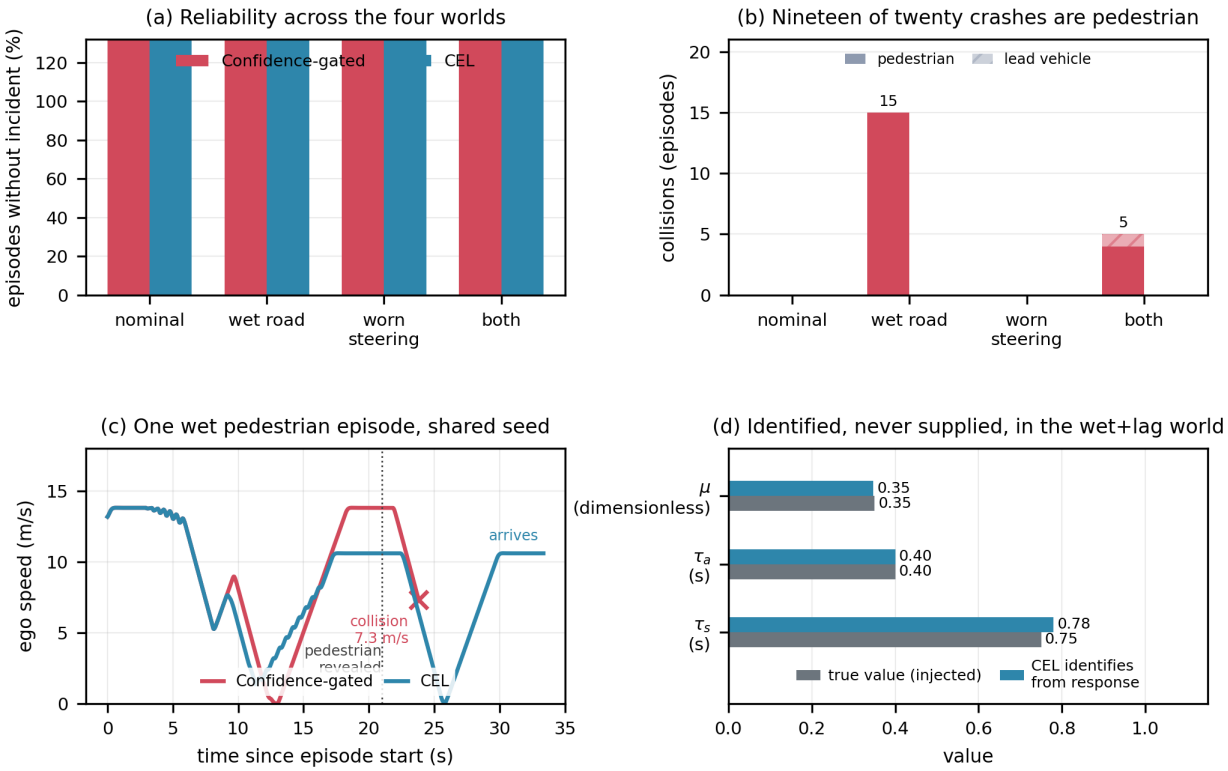


Fig. 3. The driving audit, drawn from the same records the tables are built from. (a) Episodes without incident in each of the four declared worlds; the two arms are indistinguishable in the nominal world and again under worn steering, and they separate by half the episodes in the two wet worlds. (b) Collisions, split by what was hit. Nineteen of the twenty are pedestrian; the single lead-vehicle collision is in the wet-and-worn world. Every collision in both arms is in a wet world: wetness is necessary, and the gate's confidence is not sufficient. (c) One wet pedestrian episode, the same world and the same seed for both arms (episode 35): ego speed against time, with the instant the pedestrian becomes visible marked. At that instant the gate is at 13.8 m/s and the layer at 10.6 m/s, because the layer had already priced the surface it measured. The layer's own braking command comes later—0.85 s after the reveal for the gate, 1.45 s for the layer—but it starts from a speed that a friction-limited stop can retire: it comes to rest 4.7 s after the reveal and then completes the route, while the gate is still doing 7.3 m/s where the pedestrian is. This is the section's whole argument in one trace: the difference is not reaction time, it is the speed the reaction starts from. (d) What the layer identifies in the wet-and-worn world, against the values injected: grip from saturation, the two lags from the step response. The baseline has no estimator and reports its prior in every world ($\hat{\mu} = 0.90$, $\hat{\tau}_a = 0.10$ s, $\hat{\tau}_s = 0.08$ s), which is the blindness the figure is about.

layer, of which 28 are emergency stops, and 19 of its episodes are ones where the gate braked and the blind replay arrives safely—refusals of a picture rather than of a world. The layer refuses 195 times over 32586 witness checks; 46 episodes carry at least one refusal, and every refusal is a jerk-limited ramp inside the next plan rather than a 6.0 m/s^2 step, so the lurch a passenger would feel is the one arm A makes.

Time is where the layer pays. Episode wall-clock rises from 30.7 s to 32.6 s, +1.9 s or +6.1%, and that figure is not entirely honest: 1.7 s of the baseline's mean is its twenty crashed episodes ending early. Matched on episodes that reach the goal, the cost is +0.2 s (+0.6%), and in the nominal world, where both arms arrive every time, the layer is 0.3 s *faster* (32.2 against 31.9 s) because it never brakes for a shadow. We report the flattering number and the corrected one together, since the flattering one is the one that would be quoted.

a) What this does not show.: The plant is kinematic: no tyre model, no load transfer, no perception, and the confidence is the same declared stand-in used above, a scalar function of the frame with noise and a novelty penalty rather than a trained head. One route geometry with one seeded hazard per episode means the rates estimate a mechanism, not a fleet. $\mu = 0.35$ and $\tau_s = 0.75$ s are fault injections, not measured tyres: a wet road is 0.35, and a rack that lags 0.75 s is a failing one. The magnitudes were chosen to be survivable by the layer and lethal to the gate; the ordering is the claim. Finally, the driving ledger is a real chain: 39990 registrations, 32391 grants and 195 refusals, re-walked from its genesis hash and verified, with no replay offered and none accepted, so the ordinal of Section VII holds at 10 Hz for 200 episodes.

VII. The Witness Must Be Bound in Time

We now describe a defect in the layer we built, because it is the sharpest evidence for the paper's thesis and because the first version of this experiment produced a number that was wrong in the flattering direction.

A. What the first run reported

The first full run reported a success rate of 96.0% for the layer, with **zero** witness refusals in three episodes whose bowl

finished 245 mm to 269 mm from the plate.[1] Every witness check in those episodes was granted, the ledger verified, and the chain was intact.

B. The mechanism: a replay

Fig. 4 is the sequence below, drawn: registration adds to the acceptable set, membership is never consumed, and a return to an earlier pose is therefore indistinguishable from a fresh success.

The gateway keeps one set of acceptable digests per capability, and registration adds to that set; an execution warranty is issued if the offered digest is a member, and membership is never consumed. Canonicalising the scene alone therefore made every chunk comparable with every earlier chunk of the same episode. When the bowl slipped out of the gripper during transport and fell back to the table, the observed canonical form was byte-identical to the form registered ten chunks earlier, for the close stage, when the bowl really was on the table:

```
registered, chunk 3   scene@8,22,90:table
expected, chunk 10    scene@8,22,90:gripped
observed, chunk 10    scene@8,22,90:table
```

and the gate granted a claim of gripped against an observation of table. This is a replay attack in the protocol sense: a recorded message, re-presented, accepted as fresh. The bowl returning to a previously seen pose is not an exotic event; it is what gravity does.

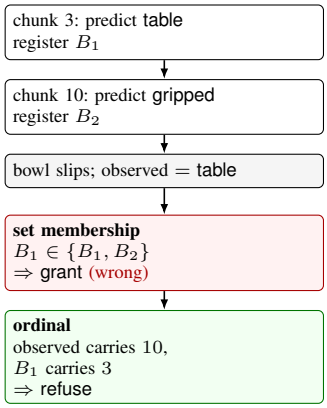


Fig. 4. The replay, and the counter that closes it. The same scene at a different attempt is a different event; a digest that binds only the scene cannot tell them apart.

C. Why hashing is not enough

A hash is an equality test on *content*. Causation is a relation between a prediction and a *particular attempt*, and two attempts may produce the same content. Formally, $H(\kappa(s)) = H(\kappa(\hat{s}_{k+1}))$ is evidence that s equals $\hat{s}_{k+1}$; it is not evidence that s is the consequence of a_k. The gap is invisible whenever states are distinguishable, which is why it survived development: the bowl on the table at the start and the bowl on the table after a slip are the same bytes, and the interesting case is exactly the one where they are.

This is the second instance of one failure in this line of work. In an earlier system the witness omitted the difficulty of the gap it witnessed, so two distinct claims shared one proof. Here the witness omits the identity of the attempt, so two distinct events share one proof. In both cases the ledger verifies, the chain passes, and the only symptom is a number that is too good.

D. The fix: a one-shot ordinal

Every registration is prefixed with a monotonically increasing ordinal, never reset, and the check for that chunk carries the same ordinal:

$$\kappa'(s, k) = \texttt{chunk}_k \,\|\, \kappa(s). \tag{3}$$

A registration is now redeemable exactly once, by the chunk that made it. A regression cannot satisfy a stale prediction because no two predictions share a key, and an episode boundary cannot alias an earlier episode's keys because the counter lives on the gateway wrapper, not on the episode. The requirement "registered before the request arrived", which a set-membership gateway only approximated, becomes a statement about the specific chunk.

The counter is a nonce, and the parallel is exact: authentication protocols have used them for decades for the same reason. A physical witness needs one because physical state is not injective in time—objects return to where they were.

E. The corrected run, and what the defect cost

After the fix, the same 200 episodes give 100.0% for the layer. The defect had made the layer *less* successful while making its ledger look *cleaner*: a false grant meant the plan continued to place a bowl it had already dropped, so it failed at the end instead of refusing in the middle. Both numbers are reported, and the pre-fix one is not archived as a result—it is recorded here as a failure, which is the only honest place for it.

VIII. Threats to Validity

a) The policy is not a foundation model.: There is no OpenVLA checkpoint in this environment, so the policy is a scripted chunked program and the "forward pass" is the declared constant c_f of Section IV. What that costs: the absolute latencies are a price list, not a measurement, which is why Equation (2) is printed. What it does not cost: the comparison is between gates over an identical policy, and the policy is open-loop within a chunk exactly as a VLA is. A better policy changes the magnitudes, not the argument.

b) The confidence head is a model of a head.: Arm A's confidence is a scalar function of frame brightness with declared noise, not a trained head. A better-calibrated head would veto less often, so our 40.5% is a property of this stand-in. But the argument is not calibration: a head consumes the frame, and the frame does not contain the plate's displacement. The *light*

[1] Quoted from the pre-fix run's console output. That run is not archived as a result: a run whose gate cannot say no is preserved as a defect record, not as a table.

channel is where this is least ambiguous, and there the head behaves exactly as a well-calibrated head should—it notices the input changed and refuses 38 times, 18 of them wrongly. Calibration moves that number; it does not move it to zero.

c) Privileged state.: The layer reads simulator ground truth. A real deployment needs a state estimator, and the witness is only as good as that estimator is accurate relative to the grid. We chose 10 mm, five times the observed settling; a perception stack with, say, 15 mm noise would need a coarser grid and would be correspondingly less sensitive. The drift audit measures the image channel separately for this reason and reports *light* as inert for this policy rather than as a capability difference.

d) One task, one seed, 100 episodes per arm.: The intervals in Table I are the honest statement of that limit. The pose-drift result ($0/45$ against $45/45$, $p < 10^{-24}$) is far outside sampling noise; the reported 100.0% should be read as "above 96.3%", not as a measurement of a rate.

e) Injected drift is parametric.: Friction, mass, light, actuator gain and a rigid displacement. Real drift includes geometry change, occlusion, sensor failure and human interference. The layer's dependence on being able to *classify* the observed state is a genuine scope limit: under occlusion the witness can be unable to answer, and the correct behaviour there is to refuse, which we do not test.

f) Prior-art coverage is not exhaustive.: The scripted sweep of arXiv and Semantic Scholar could not be completed: both agencies answered every keyword query with HTTP 429 from this host throughout the audit, so the sweep returned no records. It was replaced by a manual search of the same eight phrases over 2023–2026—*causal execution layer*, *safe VLA*, *end-to-end autonomous driving*, *world model*, *physical hallucination*, *metabolic solidification*, *safety filter* for VLA, and *runtime verification* for manipulation—with every candidate confirmed against its arXiv abstract page. A manual sweep carries no recall guarantee: an isomorphic work this vocabulary does not reach may exist, so the novelty claims here are claims about the works cited, not about the literature.

g) The chain cost is measured on a local filesystem.: Table III is a tmpfs measurement; on a network store, fsync dominates and the constant is worse. The scaling, not the constant, is the finding.

IX. Future Work

a) Sub-linear tamper-evidence.: Incremental verification with a periodically anchored Merkle root is the obvious fix for Section V-C; the open question is what guarantee survives between anchors and how much a deployment should pay for it. This is the one item on this list that is a defect rather than an extension.

b) Sequences, not chunks.: An ordinal binds one attempt. A causal trace should bind a *partial order* of attempts, so that a plan can be refused for the shape of its history and not only for the last step—for example when three re-derivations in a row are evidence of a model that is wrong about the mechanism, not unlucky.

c) Learned witnesses.: The classification used here is geometric and hand-declared. A learned witness that maps proprioception and vision to the same canonical form would remove the privileged-state assumption; the risk is the obvious one, that the witness inherits the policy's blindness.

d) When is a witness worth its price?: The layer costs a measurable per-chunk amount and returns success under drift. The ratio should be predicted before it is measured: a witness is worth buying while its price is below the drift rate times the cost of a failure, which is the same inequality that governs a compiled rule in a metabolising system. Our 54.8 ms against a 4609 ms episode is one point on that curve, not the curve.

e) Real foundation models, more tasks, more seeds.: The next experiment is this protocol with an OpenVLA checkpoint and the LIBERO-90 suite, where the relative cost changes because a real forward pass is expensive next to a witness.

X. Conclusion

A confidence gate and a physical witness answer different questions. The first asks how sure the model is; the second asks whether the world did what the model said. On a real LIBERO task, under a target that moves 100 mm after the plan is written, those questions get different answers: 0.0% against 100.0%, with the confidence baseline refusing 40.5% of the actions it vetoed and the layer refusing none that would have worked. The layer's decision latency is lower than the baseline's, because a blind retry is more expensive than a measurement; its total latency is higher, because our ledger verification is linear in the ledger, which we report as a defect.

The same question, asked of a vehicle instead of an arm, gets the same answer under a drift no camera can see: on a wet road the gate ends 15 of 30 episodes in a pedestrian collision and the layer ends none, and on a worn steering rack both arms arrive while only the layer arrives in its own lane (96% against 0% over the centre line). The layer recovers the friction and the lags from its own saturated and stepped responses, so nothing in that comparison depends on privileged state. A witness that commits to a physical consequence is portable across embodiments because it is a claim about the world—the same world in both experiments, which is more than can be said for a confidence.

The most useful result is the one that went wrong. Our first gate accepted a scene that had returned to a state it had predicted earlier, because the gate tested membership in a set of digests rather than identity of an attempt. A hash binds content; causation binds a prediction to an attempt, and the difference between those two is exactly the case where the world undoes your work. The monotonic ordinal that closes it is one line. Finding that it was needed took a run whose numbers looked good.

Reproducibility

Every number in this paper is generated. `experiments/benchmark_libero_audit.py` runs the 200 episodes and writes `libero_audit_episodes.csv`,

`libero_audit_summary.json` and the hash-linked ledger; `experiments/cel_paper/make_numbers.py` turns those into the `data/*.tex` macros and tables this document inputs. The artifact was not committed from the author's working tree; the commands and the pinned versions (`mujoco==3.2.3`, `robosuite==1.4.0`) are in the repository's `experiments/README.md`, and the defect of Section VII is `ERR-004` in its `ERRATA.md`.

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